

SAMSUZZAMAN AFROZ

PhD Researcher in Gravitational-Wave Astrophysics and Cosmology

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Education

PhD in Gravitational-Wave Astrophysics & Cosmology Tata Institute of Fundamental Research (TIFR), Mumbai, India	2022–2027 (expected)
M.Sc. in Physics Presidency University, Kolkata, India	2020–2022
B.Sc. (Hons) in Physics Jamia Millia Islamia, New Delhi, India	2017–2020

Research Interests

Gravitational-wave astrophysics and cosmology; tests of General Relativity in weak- and strong-field regimes; compact binary populations and formation channels; multimessenger cosmology with standard sirens; dark energy and cosmological parameter inference; phase-space methods for gravitational-wave source classification; dark matter and exotic compact objects.

Research Experience

Tata Institute of Fundamental Research (TIFR), Mumbai, India January 2023 – Present
PhD Research Student Advisor: Dr. Suvodip Mukherjee

- **Gravitational-Wave Astronomy:** Analyzed compact binary coalescence data from LIGO–Virgo–KAGRA to study binary black hole populations, formation channels, and their redshift evolution.
- **Multimessenger Cosmology:** Investigated constraints on the Hubble constant and the dark energy equation of state using bright and dark gravitational-wave standard sirens, in combination with electromagnetic surveys.
- **Phase-Space Methods for Source Classification:** Developed fast, high-dimensional phase-space trajectory pipelines to classify compact-object formation scenarios directly from gravitational-wave catalogs.
- **Tests of General Relativity:** Performed parametric and non-parametric tests of gravity in both weak- and strong-field regimes using current and next-generation gravitational-wave detectors.
- **Gravitational-Wave Bursts and Exotic Sources:** Studied hyperbolic encounters and unbound compact-object interactions as potential gravitational-wave burst sources and probes of dark matter and exotic compact objects.
- **Cross-Correlation with Large-Scale Structure:** Explored cross-correlation techniques between gravitational-wave events and galaxy catalogs to improve cosmological parameter inference.
- **Dark Matter with Gravitational Waves:** Investigated signatures of dark matter distributions and exotic compact objects through their imprints on gravitational-wave observations.
- **Dark Energy Consistency Tests with Cosmological Probes:** Developed a dataset-consistency framework using distance duality to assess dark-energy evolution in combined supernova (SN) and BAO observations.

Presidency University, Kolkata, India

November 2021 – July 2022

Master's Thesis

Advisor: Dr. Suchetana Chatterjee

- Studied the mass function and growth history of supermassive black holes and their astrophysical implications.

Skills

Programming & Tools: Python, C++, Fortran, Mathematica, MATLAB, HTML, Git, $\text{\LaTeX}$

Operating Systems: macOS, Linux, Windows

Cluster Computing: High-Performance Computing (HPC)

Collaborations

Former Member of the LIGO–Virgo–KAGRA Scientific Collaboration (2023–2026)

Lead Author Publications

- **Afroz, S.**, Navdha, & Mukherjee, S. (2025). *Are all Binary Black Holes Detected by LVK Following the Universal Time-Delay Distributions? Probably Not.* [arXiv:2510.06352](#) [astro-ph.HE].
- **Afroz, S.** & Mukherjee, S. (2025). *The Non Parametric Reconstruction of Binary Black Hole Mass Evolution from GWTC-4.0 Gravitational Wave Catalog.* [arXiv:2509.25356](#) [astro-ph.HE].
- **Afroz, S.** & Mukherjee, S. (2025). *Binary Black Hole Phase Space Discovers the Signature of Pair Instability Supernovae Mass Gap.* [arXiv:2509.09123](#) [astro-ph.HE].
- **Afroz, S.** & Mukherjee, S. (2025). *Gravitational Wave Burst from Bremsstrahlung in Milky Way Can Discover Sub-Solar Dark Matter in Near Future.* [arXiv:2507.22126](#) [astro-ph.CO].
- **Afroz, S.**, Mukherjee, S., & Tasinato, G. (2025). *Illuminating Dark Energy with Bright Standard Sirens from Future Detectors.* [arXiv:2507.06340](#) [astro-ph.CO].
- **Afroz, S.** & Mukherjee, S. (2025). *The Phase Space of Low-Mass Binary Compact Objects from LVK Catalog: Hints on the Chances of Different Formation Scenarios.* [arXiv:2505.22739](#) [astro-ph.HE].
- **Afroz, S.** & Mukherjee, S. (2025). *Hint towards Inconsistency between BAO and Supernovae Dataset: The Evidence of Redshift Evolving Dark Energy from DESI DR2 is Absent.* [arXiv:2504.16868](#) [astro-ph.CO].
- **Afroz, S.** & Mukherjee, S. (2024). *Multi-messenger Cosmology: A Route to Accurate Inference of Dark Energy beyond CPL Parametrization from XG Detectors.* JCAP 03 (2025) 070, [arXiv:2412.12285](#) [astro-ph.CO].
- **Afroz, S.** & Mukherjee, S. (2024). *Phase Space of Binary Black Holes from Gravitational Wave Observations to Unveil its Formation History.* Phys. Rev. D 112 (2025) 2, 023531, [arXiv:2411.07304](#) [astro-ph.HE].
- **Afroz, S.** & Mukherjee, S. (2024). *Prospect of Precision Cosmology and Testing General Relativity using Binary Black Holes–Galaxies Cross-correlation.* Mon. Not. R. Astron. Soc. 534 (2024) 2, 1283–1298, [arXiv:2407.09262](#) [astro-ph.CO].
- **Afroz, S.** & Mukherjee, S. (2024). *A Model-independent Precision Test of General Relativity using LISA Bright Standard Sirens.* JCAP 10 (2024) 100, [arXiv:2406.08791](#) [astro-ph.CO].
- **Afroz, S.** & Mukherjee, S. (2023). *A Model-independent Precision Test of General Relativity using Bright Standard Sirens from Ongoing and Upcoming Detectors.* Mon. Not. R. Astron. Soc. 530 (2024) 4, 3812–3826, [arXiv:2312.16292](#) [astro-ph.CO].
- **Full list available at:** inspirehep.net/authors/2741174

Presentations / Talks

- **A Model-Independent Precision Test of General Relativity with Gravitational Waves: A Multi-Messenger Approach**, presented at the *LVK Cosmology Working Group*, December 15, 2023.
- **Prospect of Precision Cosmology and Fundamental Physics test using Binary Black Holes- Galaxies Cross-correlation**, presented at the *LVK Cosmology Working Group*, June 28, 2024.
- **Invited Talk: A New Model-Independent Test of GR with Gravitational Waves**, presented at the *Testing Gravity with Multi-messenger Astronomy*, IIT-Bombay, July, 2024.
- **A New Model-Independent Test of GR with Gravitational Waves**, presented at the *Cosmology F2F, LVK Meeting Barcelona 2024*.
- **Phase Space of Binary Black Holes from GW Observations: A Probe to the Tale of its Formation History**, presented at the *LVK Rates & Population Working Group*, October 31, 2024.
- **Exploring New Physics and Testing General Relativity with Gravitational Waves**, presented at the *Ligo India Scientific Collaboration Meetings*, November 12, 2024.

- **Prospects of Precision Cosmology with Gravitational Waves: Blinded Mock Data Challenge and Benchmarking**, presented at the *Ligo India Scientific Collaboration Meetings*, November 19, 2024.
- **Invited Talk: Route to Accurate Inference of Dark Energy Equation of State using GWs**, presented at the *International Conference: Tensions and Anomalies on the Sky – Quest for New Physics at Cosmological Scales, Jamia Millia Islamia*, March, 2025.
- **The Phase Space of Low-Mass Binary Compact Objects from LIGO-Virgo-KAGRA Catalog**, presented at the *LVK Rates & Population Working Group*, May 15, 2025.
- **Evidence of PISN mass scale of Black Hole Mass from GWTC-4 using Phase Space Technique**, presented at the *LVK Cosmology Working Group*, August 28, 2025.
- **Non-Parametric Reconstruction of Binary Black Hole Mass Evolution from Gravitational Wave Catalog GWTC-4**, presented at the *LVK Rates & Population Working Group*, September 18, 2025.
- **Delay Time Distributions of Binary Black Hole Mergers: A GridBased Inference from GWTC-4**, presented at the *LVK Rates & Population Working Group*, September 25, 2025.
- **Invited Talk: Multi-Messenger Cosmology using bright and dark sirens**, presented at the *Cosmic Origins and the Search for New Physics*, January, IIT Madras 2026.

Professional Service / Organizing Activities

Local Organizing Committee Member, GW-EM-Nu-2023: Role of India in Multi-Messenger Astrophysics and Cosmology November 20–24, 2023

Tata Institute of Fundamental Research, Mumbai, India

- Served on the local organizing committee for a national multi-messenger astronomy conference, facilitating coordination of participants, logistics, and conference activities.

Organizing Committee Member, Young Astronomers' Meet (YAM 2025) March 18–21, 2025

Tata Institute of Fundamental Research, Mumbai, India

- Served on the organizing committee for YAM 2025, an initiative supported by the Astronomical Society of India, which provided a platform for emerging researchers in astronomy, astrophysics, and cosmology to present their work and network with peers.

Local Organizing Committee Member, GW-EM-Nu-2025: Multi-Messenger Science With Indian Facilities – Now and in the Next Decade December 1–3, 2025

Tata Institute of Fundamental Research, Mumbai, India

- Served on the local organizing committee for a national multi-messenger astronomy conference, facilitating coordination of participants, logistics, and conference activities.

Organizer, Department of Astronomy and Astrophysics Journal Club January 2024 – May 2025

Tata Institute of Fundamental Research, Mumbai, India

- Organized and coordinated a regular departmental journal club.

Schools/Lectures/Courses/Workshops

Summer School on Gravitational-Wave Astronomy 24 July–04 August 2023

International Centre for Theoretical Sciences (ICTS) – TIFR, Bengaluru, India

- Intensive training in numerical relativity and computational techniques for compact binary simulations, including the 3 + 1 decomposition of Einstein's equations, ADM and BSSN formulations, gauge conditions, and constraint stability.
- Studied relativistic hydrodynamics in curved spacetime, high-resolution shock-capturing methods, and applications to neutron star mergers and core-collapse scenarios.
- Covered well-posedness of PDEs, hyperbolic systems, and numerical methods (spectral and finite-difference) used in computational relativity.

42nd Annual Meeting of the Astronomical Society of India (ASI 2024) January 31 – February 4, 2024

Indian Institute of Science (IISc), Bengaluru, India

- Attended the annual meeting of the Astronomical Society of India and presented a *a poster* on tests of general relativity using gravitational-waves.

Summer School on Gravitational-Wave Astronomy: Continuous Gravitational Waves 01–12 July 2024

International Centre for Theoretical Sciences (ICTS) – TIFR, Bengaluru, India

- Advanced training on astrophysical and exotic sources of continuous gravitational waves in the context of LIGO–Virgo–KAGRA observations.
- Studied neutron star structure and evolution, GW emission mechanisms (mountains, r-modes), and continuous-wave search methodologies.
- Learned about superradiant instabilities and ultralight boson clouds around spinning black holes, and their GW signatures.

International Conference: Testing Gravity with Multi-messenger Astronomy July 22–24, 2024

Indian Institute of Technology Bombay, Mumbai, India

- Attended an international conference focused on Modified gravity.
- Delivered an *invited talk* on New Model-Independent Test of GR with Gravitational Waves.

Eccentricity Workshop: Challenges in Unraveling Astrophysical Eccentric Compact Binaries March 1–4, 2025

Centre for Strings, Gravitation and Cosmology (CSGC), IIT Madras, Chennai, India

- Attended a focused workshop on the dynamics, detection, and modeling of eccentric compact binaries in gravitational-wave astronomy.

International Conference: Tensions and Anomalies on the Sky – Quest for New Physics at Cosmological Scales March 6–8, 2025

Centre for Theoretical Physics, Jamia Millia Islamia (JMI) in collaboration with IUCAA, India

- Attended an international conference focused on cosmological tensions, dark matter, dark energy, and new physics beyond the standard model.
- Delivered an *invited talk* on dark energy reconstruction using gravitational-wave observations, highlighting constraints on cosmological parameters from GW standard sirens.

Discussion Meeting: The Future of Gravitational-Wave Astronomy October 27–31, 2025

International Centre for Theoretical Sciences (ICTS) – TIFR, Bengaluru, India

- Participated in a discussion meeting marking the 10th anniversary of the first gravitational-wave detection, focused on future directions in GW physics and astronomy.
- Presented *four posters and delivered a flash talk* on topics related to gravitational-wave astrophysics and cosmology.

GW Detector Characterization Workshop 15–19 December 2025

Inter-University Centre for Astronomy and Astrophysics (IUCAA), Pune, India

- Training in detector characterization for ground-based GW detectors, focusing on instrumental and environmental noise sources relevant for LIGO, Virgo, and LIGO–India.
- Hands-on experience with data-quality, glitch analysis, and signal-noise discrimination techniques.
- Exposure to detector commissioning challenges and noise mitigation strategies critical for improving GW search sensitivity.

International Conference: Cosmic Origins and the Search for New Physics January 19–23, 2026

Indian Institute of Technology Madras, Chennai, India

- Attended an international conference focused on early-universe physics, primordial gravitational waves, cosmological anomalies, and searches for new physics.
- Delivered an *invited talk* on the role of gravitational waves and galaxy surveys in probing cosmology.

Last updated: May 8, 2026